mechanics. So what is it about string theory that has allowed it to succeed thus far? We've described how Schwarz and Scherk realized, much to their surprise, that one particular string vibrational pattern had just the right properties to be the graviton particle, and how they then concluded that string theory provided a ready-made framework for merging the two theories. Historically, that is indeed how the power and promise of string theory was fortuitously realized, but as an explanation for why the string approach succeeded where all other attempts failed, it leaves us wanting. Figure 12.2 encapsulates the conflict between general relativity and quantum mechanics—on ultrashort distance (and time) scales, the frenzy of quantum uncertainty becomes so violent that the smooth geometrical model of spacetime underlying general relativity is destroyed—so the question is, How does string theory solve the problem? How does string theory calm the tumultuous fluctuations of spacetime at ultramicroscopic distances?

The main new feature of string theory is that its basic ingredient is not a point particle—a dot of no size—but instead is an object that has spatial extent. This difference is the key to string theory's success in merging gravity and quantum mechanics.

The wild frenzy depicted in Figure 12.2 arises from applying the uncertainty principle to the gravitational field; on smaller and smaller scales, the uncertainty principle implies that fluctuations in the gravitational field get larger and larger. On such extremely tiny distance scales, though, we should describe the gravitational field in terms of its fundamental constituents, gravitons, much as on molecular scales we should describe water in terms of H_2O molecules. In this language, the frenzied gravitational field undulations should be thought of as large numbers of gravitons wildly flitting this way and that, like bits of dirt and dust caught up in a ferocious tornado. Now, if gravitons were point particles (as envisioned in all earlier, failed attempts to merge general relativity and quantum mechanics), Figure 12.2 would accurately reflect their collective behavior: ever shorter distance scales, ever greater agitation. But string theory changes this conclusion.

In string theory, each graviton is a vibrating string—something that is not a point, but instead is roughly a Planck length (10^{-33} centimeters) in